

6(a)	- p.d. across resistor = p.d. across capacitor - current (in resistor) proportional to p.d. across it - current causes capacitor to lose charge - charge (on capacitor) proportional to p.d. so p.d. decreases <i>Any two points, 1 mark each</i>	B2
	rate of change of p.d. decreases as p.d. decreases	B1
6(b)	$Q_0 = 0.90 \text{ mC}$ and at $t =$ one time constant, $Q = Q_0 \exp(-1)$	B1
	at $t =$ one time constant, $Q = 0.90 \exp(-1) = 0.33 \text{ mC}$	M1
	evidence of graph reading: when $Q = 0.33 \text{ mC}$, $t = 5.5 \text{ s}$	A1
	or	
	evidence of two correct sets of readings for Q and t from the graph	(B1)
	correct substitution of Q and t values into $Q_2 = Q_1 \exp[(t_1 - t_2) / \tau]$	(M1)
	calculation to give $\tau = 5.5 \text{ s}$	(A1)
	or	
	read-off of half-life as 3.75 s	(B1)
	use of $Q = Q_0 \exp(-t / \tau)$ to show that $\tau = \text{half-life} / \ln 2$	(M1)
	$\tau = 3.75 / \ln 2 = 5.4 \text{ s}$	(A1)

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Question	Answer	Marks
6(b)	or	
	tangent drawn on $Q-t$ graph and value of Q at exact same time as tangent read from graph	(M1)
	gradient of tangent correctly calculated	(A1)
	$\tau = Q / \text{gradient}$ used to correctly calculate a value for τ as 5.5 s	(A1)
6(c)(i)	$C = Q / V$	C1
	$= [(0.90 \times 10^{-3}) / 7.5] = 1.2 \times 10^{-4} \text{ C}$	A1
	$= 120 \mu\text{F}$	
6(c)(ii)	$R = \tau / C$	C1
	$= 5.5 / (1.2 \times 10^{-4}) (= 45\,800 \Omega)$	A1
	$= 46 \text{ k}\Omega$	